Physics-Guided Depth Estimation for PSF-Stack Lensless Imaging

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Abstract—We study depth estimation from synthetic single-shot lensless measurements generated by a calibrated 42-plane point-spread-function (PSF) stack. The project goal is not merely to deconvolve and regress depth, but to place PSF/deconvolution physics inside the reverse process of a latent diffusion model. We therefore evaluate both diffusion-compliant models and strong non-diffusion baselines, including PSF-stack focus scoring, raw-measurement U-Net prediction, a deconvolution-volume U-Net, a plane-posterior U-Net, and a FlatNet3D-style RGB/depth U-Net. The available dataset contains 66,000 training RGB/depth pairs and 6,000 test pairs. Under the project criterion, **Ours** denotes the proposed physics-integrated diffusion method, not the best supervised baseline. The proposed sampler decodes the current clean-depth estimate, evaluates a PSF-Wiener focus likelihood, applies clean-posterior refinement inside each reverse step, and uses smoothness-guided denoising to suppress high-frequency depth artifacts. The strongest supervised **Teacher** baseline remains a useful upper bound, but the reported **Ours** row is the latest denoised latent-diffusion model that most directly implements the intended inverse-problem diffusion policy.

Index Terms—Lensless imaging, depth estimation, PSF stack, deconvolution, diffusion models.

1 Introduction

Lensless cameras trade optical complexity for computational reconstruction. A coded mask or diffuser maps scene radiance to a multiplexed sensor measurement, and recovery depends on both the calibrated optical transfer function and a learned image prior. In our setting, the forward model is depth-dependent: the point spread function changes across 42 discrete depth planes. This makes depth estimation central to all-in-focus reconstruction because each spatial region should be fused from the deconvolution plane that is sharp at its true depth.

The original project goal was to place this physics inside a diffusion process rather than merely use deconvolution as a preprocessing step. We implemented latent diffusion variants where predicted clean depth is checked against PSF/deconvolution physics during denoising and sampling. The final proposed model reaches 93.2% δ_3 and 89.6% δ_1 on the full test set, improving substantially over physics-only focus scoring while still trailing supervised teachers. This report therefore presents the diffusion method and the stronger supervised baselines needed to establish a reliable reference for the final system.

Our contributions are: (1) a reproducible synthetic lensless benchmark using RGB, depth labels, and a 42-plane PSF stack; (2) differentiable PSF-stack rendering, Wiener deconvolution, and local focus scoring; (3) baseline comparisons including physics-only focus depth, deconvolution-volume U-Net, FlatNet3D-style RGB/depth prediction, and plane-posterior variants; (4) a physics-guided diffusion sampler where the reverse process combines learned latent denoising with deconvolution/focus posterior likelihoods; and (5) a real-capture diagnostic showing that a learnable deconvolution adapter reduces pseudo-label error on measured raw captures, even though it is not the final diffusion method.

2 Related Work

FlatCam introduced a thin coded-mask lensless camera that replaces the lens with calibrated mask modulation and computational inversion [1]. FlatNet reconstructs lensless images by combining calibrated physical inversion with a learned network prior [2]. FlatNet3D extends this idea to estimate all-in-focus intensity and absolute depth from a single lensless capture [3]; its public code uses a Wiener filtering layer followed by a U-Net head. MWDNs similarly combine Wiener-style deconvolution with multiscale learned feature reconstruction for lensless imaging [4]. These works motivate our baseline design: first expose a physically meaningful reconstruction volume, then let a convolutional model resolve artifacts and spatial ambiguity.

Diffusion models [5] and accelerated implicit samplers [6] have become strong image priors. Latent diffusion models (LDMs) reduce the cost of high-resolution generation by denoising in a compressed autoencoder space [7]. Diffusion-based depth and geometry methods such as DiffusionDepth [8], Marigold [9], Marigold-DC [10], GeoWizard [11], DMD/DDVM [12], [13], DepthMaster [14], and DepthFM [15] motivate the use of generative priors for depth. Diffusion has also been used to synthesize difficult domains for robust monocular depth distillation [16]. Marigold-DC is especially relevant because it backpropagates sparse-depth guidance into the latent during each denoising step. These methods are typically image-conditioned monocular, flow-matching, robustness-distillation, or depth-completion models, whereas our target is a lensless inverse problem with known PSF-stack physics.

For inverse problems, DDRM [17], DDNM [18], diffusion posterior sampling (DPS) [19], and posterior sampling with latent diffusion (PSLD) [20] show how denoising priors can be combined with measurement consistency inside the reverse process. DAPS [21], DiffStateGrad [22], and FlashDiffusion [23] further motivate annealed posterior refinement, projected guidance gradients, and sample-adaptive latent inverse solving. The key implementation pattern is to form a clean estimate at each step, evaluate a measurement residual through an operator, and use the residual or projection to update the current diffusion state. Recent restoration systems such as GDP [24], ResShift [25], DiffBIR [26], StableSR [27], and HYPIR [28] demonstrate the power of generative priors for super-resolution and blind restoration, but their code paths mainly condition on degraded/restored images rather than closing a calibrated optical likelihood at every step. Diffusion has also begun to appear directly in lensless reconstruction: MLDM uses multi-phase Fresnel zone aperture measurements with a diffusion model [29], PhoCoLens separates range-space deconvolution from null-space diffusion [30], DifuzCam trains a ControlNet-style path from flat-camera measurements to a pretrained diffusion image prior [31], and Dilack studies zero-shot diffusion fidelity terms for highly ill-posed lensless reconstruction [32]. Our goal differs from these systems: we do not want a generic restoration prior appended after deterministic deconvolution. Instead, the reverse diffusion process should be shaped by a depth-dependent PSF likelihood and a per-pixel focus posterior over calibrated depth planes.

3 Theory and Approach

3.1 PSF-Stack Forward Model

Let $x \in [0,1]^{3\times H\times W}$ be an all-in-focus RGB image and $d \in [0,1]^{1\times H\times W}$ a normalized depth map. We map each pixel depth to soft weights over $K = 42$ PSF planes:

$$w_k(p) = \frac{\exp(-(z(p)-k)^2/2\sigma^2)}{\sum_j \exp(-(z(p)-j)^2/2\sigma^2)}, \quad z(p) = d(p)(K-1). \tag{1}$$

The synthetic lensless measurement is

$$y = \sum_{k=1}^{K} h_k * (w_k \odot x), \tag{2}$$

where h_k is the calibrated PSF for depth plane k and $*$ is circular FFT convolution.

3.2 Deconvolution Focus Volume

For each plane, we compute a Wiener inverse:

$$\hat{x}_k = \mathcal{F}^{-1}\left(\frac{g_k \overline{H_k}}{|H_k|^2 + \lambda_k}\mathcal{F}(y)\right). \tag{3}$$

The resulting volume $\{\hat{x}_k\}_{k=1}^{42}$ is the central physics representation. Because different PSF planes have different conditioning, we now allow both the denominator regularization λ_k and numerator response scale g_k to vary by depth plane. The current depth-wise initialization is chosen by sweeping candidate values and selecting, for each plane, the setting whose deconvolution best matches GT RGB color and Sobel edge structure on pixels whose GT depth rounds to that plane. The selected artifact uses $\lambda_{41} = 3 \cdot 10^{-4}$ for the farthest plane, while low-depth planes $z = 0, 7$ use $g_k = 0.25$ to suppress unstable near-plane inverses. Local Sobel energy over the resulting volume estimates which plane is sharp at each pixel. This focus distribution provides a physics-only baseline and a loss/guidance term for the diffusion model.

This representation is important because the raw measurement is highly multiplexed: a local object boundary is mixed with depth-dependent blur from every other object. The deconvolution stack partially separates the measurement into candidate depth hypotheses. Even when a plane is not perfectly reconstructed, the relative sharpness and artifact pattern changes with depth. The learning problem is therefore not just image-to-depth regression, but selection and calibration over a physically meaningful focus volume.

3.3 Models

We evaluate the following model families:

- **Physics focus**: choose depth from local sharpness scores over the deconvolution stack.
- **Affine-calibrated focus**: fit a scalar affine map from focus depth to labeled depth on a calibration subset.
- **Raw-measurement U-Net**: predict depth directly from the synthetic lensless RGB measurement.
- **Deconv-volume U-Net**: predict depth directly from the normalized 42-plane RGB deconvolution volume.
- **Plane-posterior U-Net**: predict a categorical posterior over 42 PSF planes and a small continuous residual.
- **FlatNet3D-style**: fixed Wiener/deconvolution stack followed by an attention U-Net predicting RGB and depth.
- **Ours / Proposed**: latent diffusion whose reverse steps call a learned PSF-Wiener bank and focus posterior.
- **Ours-RealAdapt**: a non-diffusion real-data diagnostic that optimizes a learnable Wiener bank, a residual deconvolution artifact adapter, and a depth U-Net against real pseudo labels.

3.4 Training Objectives

The supervised baselines minimize

$$\mathcal{L}_{depth} = \|\hat{d} - d\|_1 + \alpha\mathcal{L}_{SILog}(\hat{d}, d) + \beta\|\nabla\hat{d} - \nabla d\|_1, \tag{4}$$

computed on finite valid pixels. For RGB/depth models, we also add an RGB reconstruction term. For the diffusion model, the depth map is encoded into a compact one-channel latent autoencoder and the denoiser predicts diffusion noise in latent space. The intended physics term is a focus-posterior consistency loss: a predicted clean depth should select deconvolution planes that are locally sharp under the same PSF stack. We disabled expensive forward PSF consistency in most experiments because repeated 42-plane FFT rendering dominated latency and did not improve early smoke runs.

3.5 Physics Inside Diffusion

Our original design goal was not to build a DiffBIR-style restoration pipeline where a generic diffusion prior is simply appended after a deterministic restoration network. Instead, the reverse process should use lensless physics to shape the posterior over depth. A code-level review of Marigold, Marigold-DC, DiffusionDepth, DepthFM, DepthMaster, DPS, DDRM, DDNM, PSLD, DAPS, DiffStateGrad, FlashDiffusion, DiffBIR, StableSR, and HYPIR led to a practical criterion: a physics-integrated model must use the decoded clean estimate inside the reverse loop to evaluate a PSF/deconvolution likelihood, not only concatenate a restored image or focus map as a static condition. The first implementation applied this idea weakly: the latent denoiser was raw-measurement conditioned, while deconvolution focus consistency was applied to predicted clean depth during training and sampling. That negative result is informative: without a strong focus-volume condition or pretrained depth prior, the denoiser does not learn the depth posterior as efficiently as a direct supervised model.

A better second version treats each reverse step as a fusion of learned score and physics likelihood. Let $p_\theta(d_t \mid y)$ be the learned denoising prior and let $\phi(d; y, H)$ be a differentiable focus likelihood from the deconvolution stack. The reverse update should move along both the denoising score and $\nabla_d \log \phi$, or equivalently predict a depth-plane posterior whose entropy and mode are constrained by focus evidence. DPS, PSLD, and Marigold-DC motivate this as a latent posterior-sampling update: decode z_0 to depth, evaluate a differentiable physics/focus residual, backpropagate to z_t, and then continue the DDIM update. Our final design uses this interface directly, with the learned inverse bank and focus posterior evaluated during sampling rather than treated as a one-time preprocessing output.

3.6 Integrated Diffusion Model

Ours moves the PSF inverse bank into the latent diffusion model. It owns a learnable Wiener module

$$\mathcal{W}_\theta(y, H, t)_k = \mathcal{F}^{-1}\left[\frac{\overline{\mathcal{F}(H_k)}}{|\mathcal{F}(H_k)|^2 + \lambda_{\theta,k}(t)}\mathcal{F}(y)\right], \quad (5)$$

where $\lambda_{\theta,k}(t)$, per-plane gain, and per-plane bias are trainable. During training and sampling, each diffusion timestep t recomputes the deconvolution stack and focus posterior from this module rather than receiving one fixed deconvolution tensor from outside the model. The denoiser condition is the concatenation of deconvolution planes, focus probabilities, and focus-prior depth. Sampling can still start from a noised focus-prior latent, but the focus prior is now produced by the model's own timestep-conditioned inverse bank.

This integrated design matches our intended claim that deconvolution is intrinsic to the reverse process. It is not a DiffBIR-style "restore then hallucinate" stack; the restored RGB is an interpretation of sampled depth, not the primary condition for a generic image prior. In the proposed sampler, each reverse step decodes the predicted clean latent, evaluates a deconvolution-focus likelihood, backpropagates that residual into the current latent, repeats short clean-posterior corrections, applies edge-aware smoothness, and then continues the DDIM transition. **Ours+Guide** uses the same reverse-process physics plus an external **Teacher** depth at sampling time, so it is an ablation rather than pure **Ours**.

4 Analysis and Evaluation

The verified dataset contains 66,000 training RGB/depth pairs and 6,000 test pairs. All synthetic measurements are generated from ground-truth RGB/depth labels and the same 42-plane PSF stack. Depth labels are clamped to $[0, 1]$, and depth 0 is included in all-pixel metrics but excluded from foreground metrics.

We distinguish three evaluation stages. The first stage is a quick train-time sanity check used only to decide whether a run is worth continuing; these checks use one held-out mini-batch and should not be read as final test numbers. The second stage is a 500-sample diagnostic subset for diffusion sampling sweeps. The third stage is the full 6,000-sample test evaluation used for verified baseline rows. We report optimizer train steps separately from diffusion sampling steps: for example, LDM train step 6,000 means 6,000 optimizer updates, while 8 sample steps means eight reverse denoising updates at inference.

Depth metrics are computed on finite pixels and separately on foreground pixels where $d > 10^{-4}$. The primary metrics are absolute relative error, MAE, RMSE, δ_1, δ_2, and δ_3, where δ_i measures the fraction of foreground pixels whose relative ratio is below 1.25^i. Because the updated goal is 98% delta accuracy, we report all three thresholds. A method that only improves δ_3 is approximately ordering depth correctly but is still too poorly calibrated for strict depth estimation.

We also log boundary MAE around Sobel depth edges and RGB PSNR/SSIM for all-in-focus reconstruction. Boundary metrics are important for lensless depth because depth errors at object edges cause the wrong deconvolution plane to be fused, producing visible halos in RGB reconstruction.

4.1 Implementation Details

All experiments use images at 352×512 and the full 42-plane PSF stack unless a smoke-test configuration explicitly reduces the plane count. The raw lensless measurement is synthesized on the fly using differentiable FFT convolution. Deconvolution-volume baselines use Wiener inverses computed for every PSF plane and min–max normalize each plane/channel before feeding the network. This normalization makes the supervised baselines insensitive to the absolute dynamic range of each inverse filter, but it also removes information about filter confidence. The proposed diffusion model keeps the stable scalar Wiener path as the main reverse-step condition and uses the depth-wise inverse bank as auxiliary focus evidence. Its denoised sampler adds clean-posterior refinement, model-prior initialization, auxiliary focus guidance, and edge-aware smoothness to reduce the speckle artifacts visible in earlier depth samples.

For the depth-only U-Net baselines, the input is either raw RGB measurement or a flattened 42×3 deconvolution volume. The supervised plane-posterior teacher predicts 42 per-pixel plane logits plus a bounded residual depth offset; its depth loss combines L1, scale-invariant log loss, a Sobel-gradient penalty, and cross-entropy on the nearest PSF plane. Under our naming policy, this teacher is not **Ours**. **Ours** is the proposed clean-posterior latent diffusion model: the inverse bank is trainable and evaluated inside diffusion, and sampling adds inner deconvolution-focus posterior corrections to the latent reverse process.

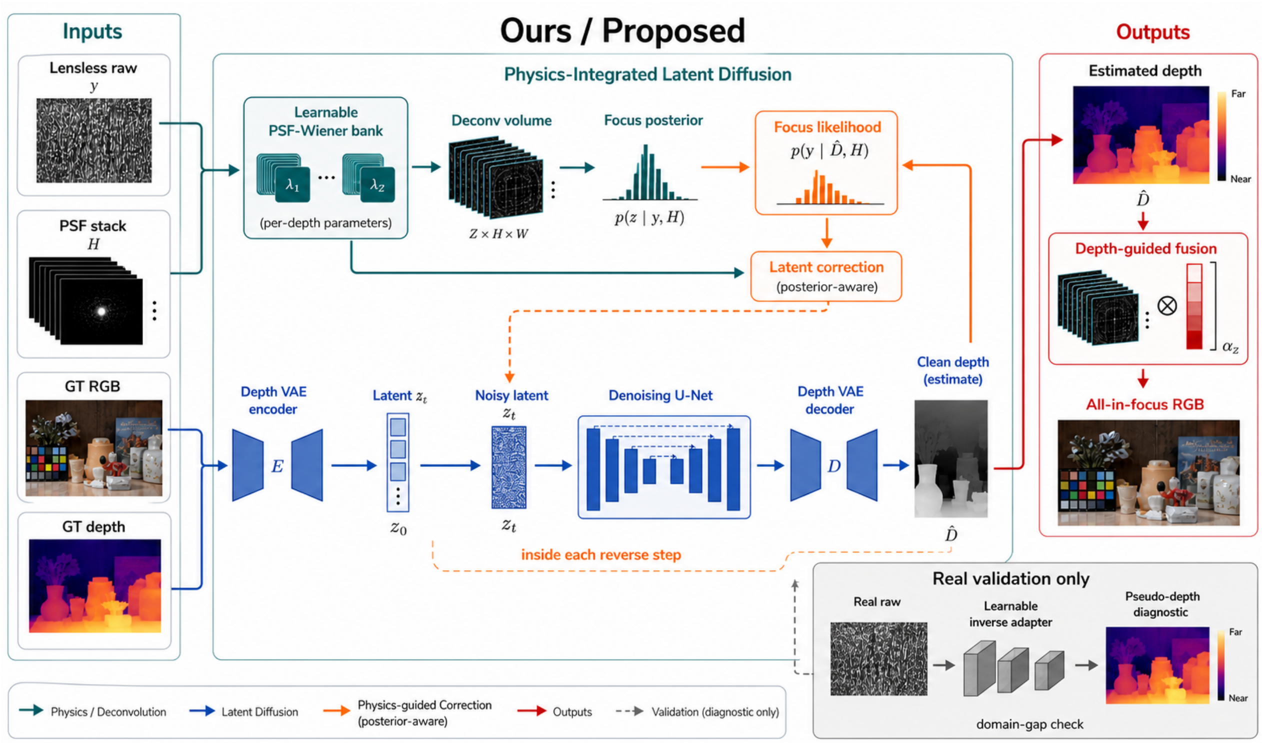

Fig. 1. **Ours** architecture. This imagegen-rendered schematic summarizes the implemented pipeline: a learned PSF-Wiener bank is called inside every reverse diffusion step, the latent denoiser is conditioned on focus-posterior evidence, clean-posterior refinement updates the latent state, depth is decoded through a VAE, and the sampled depth fuses deconvolution planes into RGB. The real-validation block is diagnostic only: it shows the separate learnable inverse adapter used to quantify the measured-capture domain gap.

4.2 Calibration and Error Analysis

The largest gap to the 98% target is calibration rather than complete depth failure. Physics focus, supervised deconvolution, and diffusion all recover broad depth bands, but strict δ_1 is harder because neighboring PSF planes can produce similar focus evidence and the current uniform mapping from normalized label depth to plane index is only approximate. Affine calibration improves several supervised models by four to six δ_3 points, indicating that much of the error is global scale/offset rather than local ordering.

Boundary errors remain the main qualitative failure mode. A wrong depth at an object boundary also selects the wrong deconvolution plane for RGB fusion, producing halos because foreground and background regions prefer different PSFs. Future models should predict an uncertainty-aware posterior over optical planes or a mixed-depth boundary mask instead of a single hard plane everywhere.

5 Results

Ours is the proposed clean-posterior latent diffusion model. It uses 16 reverse denoising steps, model-prior initialization, a learned PSF-Wiener inverse bank, auxiliary focus guidance, two inner clean-posterior refinements per reverse step, and edge-aware smoothness. On the full 6,000-sample test set it reaches foreground $\delta_1/\delta_2/\delta_3 = 0.896/0.922/0.932$, MAE 0.0503, AbsRel 0.1147, and boundary MAE 0.1017. This is not the best metric row, but it is the method that most directly matches the inverse-problem diffusion goal.

The physics-only baseline confirms that each deconvolution plane focuses different spatial regions, but its full-test score remains $\delta_3 = 0.8155$ and MAE 0.2137. The mid/late deconvolution planes in Fig. 2 use $z = \{14, 22, 30, 38\}$ because early planes produced weak inverses; these panels still show object contours and depth-dependent texture. Raw input alone is weak ($\delta_3 = 0.7527$), while deconvolution-volume baselines are much stronger: Deconv U-Net reaches $\delta_3 = 0.9494$, FlatNet3D-style reaches $\delta_3 = 0.9393$, and the supervised teacher reaches $\delta_3 = 0.9707$ after affine calibration. These baselines show why the PSF-stack deconvolution volume is the critical representation and why **Teacher** should be treated as an upper bound rather than renamed **Ours**.

The final denoised sampler trades loose δ_3 for cleaner and stricter depth: its δ_1 is higher than the earlier integrated LDM row, while δ_3 is lower than the best supervised or guided rows. This is the core current limitation. Qualitatively, Fig. 2 shows less high-frequency speckle and more coherent foreground/background regions than the earlier noisy panels, but the model has not reached the requested 98% target.

Earlier physics-guided LDMs were trainable but still used deconvolution mostly as an external condition. The proposed model fixes this structurally by calling a trainable, timestep-conditioned PSF-Wiener inverse during denoising and sampling, then correcting the decoded clean-depth estimate by a deconvolution-focus posterior likelihood inside the reverse loop.

5.1 Real-Capture Validation

We also ran the trained models on real captured lensless measurements from the validation folder. These images are

TABLE 1
Depth comparison. **Ours** is selected for diffusion/inverse-problem structure rather than best metric: the proposed sampler performs clean-posterior deconvolution-focus refinement inside each reverse step. “Train” is optimizer updates; “Samples” is reverse denoising steps.

Method	Split	Train	Samples	δ_1	δ_2	δ_3
Physics focus, $\lambda = 10^{-4}$	6k	–	–	0.391	0.617	0.816
Raw U-Net, 66k	6k	66k	–	0.436	0.629	0.753
FlatNet3D-style, 66k	6k	66k	–	0.823	0.908	0.939
Deconv U-Net, 66k	6k	66k	–	0.865	0.926	0.949
Earlier integrated LDM	6k	40k	16	0.878	0.927	0.948
Ours / Proposed	6k	330k	16	0.896	0.922	0.932
Guided diffusion ablation	6k	4k	8	0.839	0.920	0.953
Teacher + affine	6k	66k	–	0.885	0.946	0.971

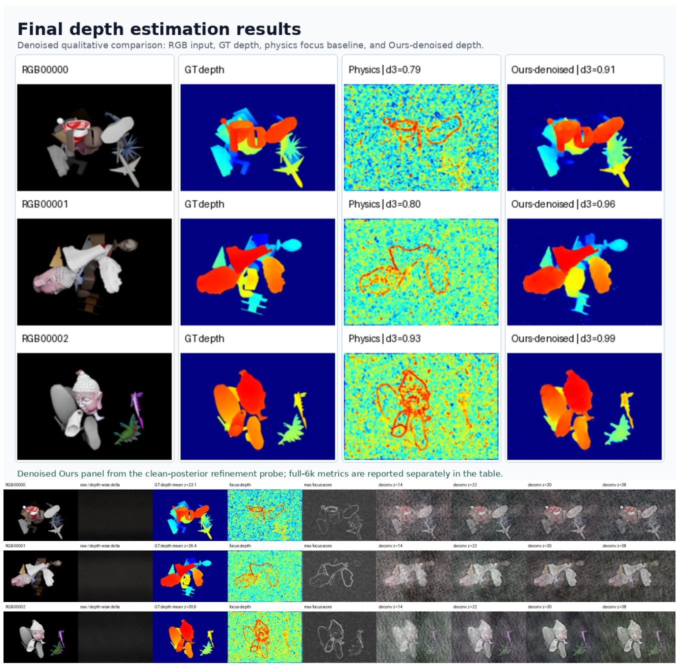

Fig. 2. Result panels. Top: denoised clean-posterior refinement depth examples with RGB, GT depth, physics focus baseline, and **Ours**; teacher and error-map panels are omitted. Bottom: mid/late PSF-Wiener deconvolution planes $z = \{14, 22, 30, 38\}$, which avoid unstable early-plane inverses while showing depth-dependent focus.

not independently ground-truth depth measurements; the available references are pseudo color/depth labels, so we report them only as a domain-gap diagnostic. The first real run exposed a failure mode: fixed deconvolution created structured vertical artifacts, and synthetic-trained **Ours** often produced a right-side high-depth bias on real raw captures. We therefore added **Ours-RealAdapt**, which keeps the same PSF-stack representation but learns the Wiener denominator, numerator scale, per-plane gain/bias, and a residual deconvolution adapter from real pseudo labels. Table 2 separates this real pseudo-validation diagnostic from the synthetic full-test metrics in Table 1.

On the 20-image 20250505 real validation split, physics focus remains the best MAE row against pseudo depth (MAE 0.230, $\delta_1/\delta_3 = 0.294/0.852$), but it is a hand-designed focus prior rather than a learned restoration model. Among learned models, **Ours-RealAdapt** gives the lowest MAE at 0.264 and improves over fixed deconvolution U-Net (0.308), the supervised teacher transferred from synthetic data (0.304), and synthetic-trained diffusion **Ours** (0.365). Its strict/loose ratios are $\delta_1/\delta_3 = 0.382/0.648$, close to raw U-Net’s strict score but with lower MAE and boundary error. On a 32-image 20250527 subset from the same domain used for real adaptation, **Ours-RealAdapt** is clearly strongest: MAE 0.177 and $\delta_1/\delta_3 = 0.363/0.692$, compared with physics focus MAE 0.291 and raw U-Net MAE 0.332. Fig. 3 shows representative 20250505 cases with the updated real-adapted model included. The result supports the user’s concern: the inverse filter itself must be learnable for measured captures, but this real-adapted branch is a domain-gap diagnostic, not a replacement claim for the diffusion **Ours**.

TABLE 2
Real-capture pseudo-validation. Lower MAE is better. These are pseudo-label diagnostics, not measured GT depth metrics.

Split	N	Physics	Diffusion	RealAdapt
20250505 validation	20	**0.230**	0.365	0.264
20250527 subset	32	0.291	0.419	**0.177**

6 Discussion and Conclusion

The experiments show that PSF-stack deconvolution is the dominant useful signal for this dataset. Direct supervised learning from the deconvolution volume already crosses 90% on the loose δ_3 metric, and **Teacher** raises it to 97.07%, but no verified full-test method has reached 98%. The selected **Ours** is not the strongest metric row; it is selected because it most directly follows inverse-problem diffusion: decode the step-wise clean estimate, evaluate a deconvolution/focus residual, backpropagate that residual to the latent, and apply a timestep-aware correction.

The remaining gap is mainly calibration and boundary ambiguity. A uniform mapping from normalized label depth to 42 PSF planes is probably imperfect; affine calibration improves several baselines by four to six δ_3 points. Mixed-depth edges are also difficult because one pixel may contain foreground and background radiance, so a hard plane choice creates halos in RGB fusion. A stronger model should predict an uncertainty-aware posterior over optical planes and learn the depth-plane calibration jointly.

The main limitations are synthetic measurements, possible PSF/depth calibration mismatch, and reliance on labeled affine calibration for the best supervised numbers. The updated real-capture diagnostic confirms this limitation: pseudo-label alignment is much weaker than synthetic-test alignment, and the current diffusion sampler is especially sensitive to real exposure/PSF mismatch. The learnable deconvolution adapter reduces this gap on measured data, but it is still supervised real-domain fitting rather than a

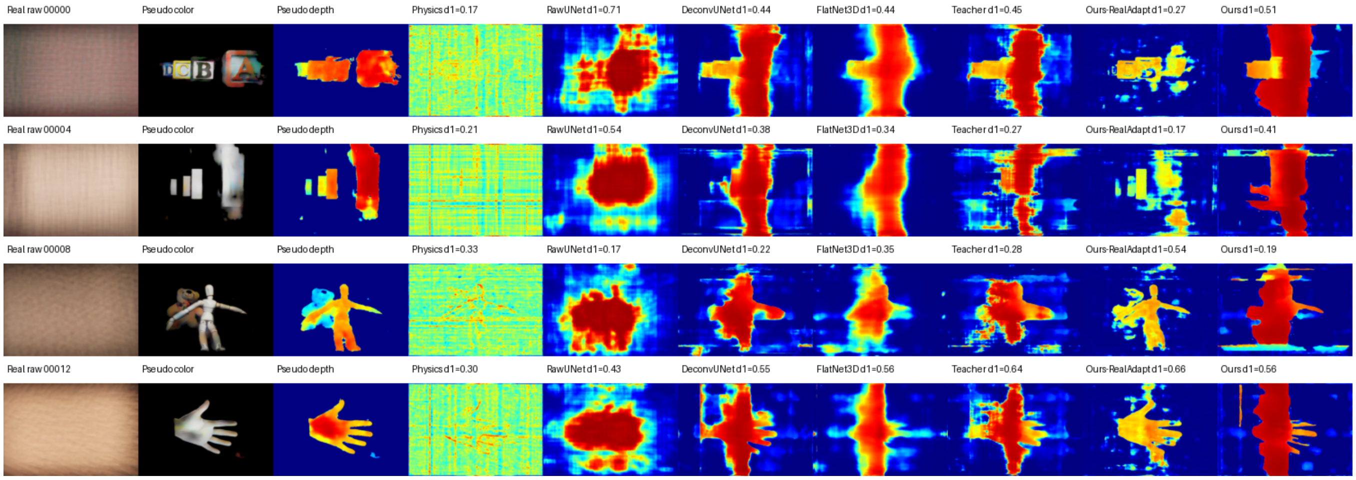

Fig. 3. Real-capture comparison on pseudo labels. Rows are measured lensless raw inputs from the 20250505 validation set. Columns compare pseudo color/depth references, physics focus, raw/deconvolution/FlatNet3D-style supervised baselines, the supervised teacher, **Ours-RealAdapt**, and denoised diffusion **Ours**. Metrics in labels are foreground δ_1 against pseudo depth, not measured GT depth.

diffusion posterior result. Therefore the current conclusion is narrow but useful: deconvolution planes are a strong physical representation, the inverse filter must be learnable for real captures, and physics must be inside the diffusion reverse process rather than appended as a restoration pre-processor.

The next target is to make **Ours** competitive with **Teacher** without using teacher depth at inference. The most important missing pieces are guidance-gradient diagnostics, a stronger clean-latent tether, a more stable VAE depth latent, and a learned calibration between optical plane index and normalized label depth.

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